Bayesian Modelling: Foundations and Conjugate Models

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[0] C. P. Robert. The Bayesian Choice. Springer, 2007.

Overview

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Frequentist statistics and the likelihood

Likelihood inference

- Let $\mathbf{X}_{1:n} := (X_1, \ldots, X_n)$ be i.i.d. random variables with density $f(x|\theta)$. Let $\mathbf{x}_{1:n} := x_1, \ldots, x_n$ be a realised sample.
- The likelihood and log-likelihood are

$$L(\theta|\mathbf{x}_{1:n}) = \prod_{i=1}^{n} f(x_i|\theta),$$

$$\ell(\theta|\mathbf{x}_{1:n}) = \sum_{i=1}^{n} \log f(x_i|\theta).$$

- The maximum likelihood estimator, $\widehat{\theta}_{\mathsf{MLE}}$, is the maximiser of $L(\cdot|\mathbf{x}_{1:n})$, or equivalently $\ell(\cdot|\mathbf{x}_{1:n})$.

Example: Poisson model

- Suppose $X_i \sim \text{Pois}(\lambda)$, i.e.

$$f(x|\lambda) = \mathbb{P}_\lambda(X_i = x) = e^{-\lambda}\frac{\lambda^x}{x!}.$$

- Up to constants independent of λ,

$$\ell(\lambda|\mathbf{x}_{1:n}) = \log(\lambda)\sum_{i=1}^{n} x_i - n\lambda,$$

$$\ell'(\lambda|\mathbf{x}_{1:n}) = \frac{1}{\lambda}\sum_{i=1}^{n} x_i - n = 0,$$

$$\widehat{\lambda}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} x_i =: \bar{x}.$$

- Recall that $X \sim \text{Pois}(\lambda)$ implies $\mathbb{E}[X] = \lambda$.

Likelihood inference

- The score function and Fisher information are, respectively,

$$S(\theta) := \frac{\partial \ell(\theta|\mathbf{x}_{1:n})}{\partial \theta} = \ell'(\theta|\mathbf{x}_{1:n}),$$
$$I(\theta) := -\frac{\partial^2 \ell(\theta|\mathbf{x}_{1:n})}{\partial \theta^2} = -S'(\theta) = -\ell''(\theta|\mathbf{x}_{1:n}).$$

- $I(\widehat{\theta}_{\text{MLE}})$ is the observed Fisher information.
- Under regularity conditions, as $n \to \infty$,

$$\widehat{\theta}_{\text{MLE}} \sim N(\theta, I^{-1}(\widehat{\theta}_{\text{MLE}})),$$

and for $\alpha \in (0, 1)$ we obtain level-α confidence intervals

$$\left[\widehat{\theta}_{\text{MLE}} \pm z_{1-\alpha/2}\sqrt{I^{-1}(\widehat{\theta}_{\text{MLE}})}\right].$$

Hypothesis testing

- A Wald test for $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$ is based on

$$T := (\widehat{\theta}_{\mathrm{MLE}} - \theta_0)\sqrt{I(\widehat{\theta}_{\mathrm{MLE}})}.$$

- Under H_0, as $n \to \infty$ we have

$$T \sim N(0, 1).$$

- Hence we reject H_0 at level $\alpha \in (0, 1)$ if

$$|T| > z_{1-\alpha/2}.$$

- Duality between the hypothesis test and confidence interval: H_0 is rejected at level α if and only if the same the same confidence interval does not contain $\widehat{\theta}_{\mathrm{MLE}}$.

Frequentist statistics

- The meaning of

$$\mathsf{CI}(\alpha) := \left[\widehat{\theta}_{\mathsf{MLE}} \pm z_{1-\alpha/2}\sqrt{I^{-1}(\widehat{\theta}_{\mathsf{MLE}})}\right]$$

  being a level-α confidence interval is

$$\mathbb{P}(\text{independent replicate } \widehat{\theta}_{\mathsf{MLE}} \in \mathsf{CI}(\alpha)) = 1 - \alpha.$$

- It is not

$$\mathbb{P}(\theta \in \mathsf{CI}(\alpha)) = 1 - \alpha.$$

- The latter is unknowable, and *nonsensical* because nothing inside $\mathbb{P}(\cdot)$ is random under frequentist assumptions.

Example: Poisson model continued

- Consider $H_0 : \lambda = \lambda_0$ vs. $H_1 : \lambda \neq \lambda_0$.
- Recall $\widehat{\lambda}_{\text{MLE}} = \bar{x}$.
- The observed fisher information is

$$-\ell''(\lambda|\mathbf{x}_{1:n})\big|_{\lambda=\widehat{\lambda}_{\text{MLE}}} = \frac{n\bar{x}}{\widehat{\lambda}^2_{\text{MLE}}} = \frac{n}{\bar{x}}.$$

- The Wald test statistic and its asymptotic distribution are

$$T = (\bar{x} - \lambda_0)\sqrt{\frac{n}{\bar{x}}} \sim N(0, 1),$$

and an approximate confidence interval for λ is

$$\left[\bar{x} \pm z_{1-\alpha/2}\sqrt{\frac{\bar{x}}{n}}\right].$$

Newton's method

- Often $S(\theta) = 0$ can't be solved analytically.
- Start with an initial guess $\theta^{(0)}$ and consider the Taylor expansion (neglecting remainder terms)

$$\begin{aligned} g(\theta) &:= S(\theta^{(0)}) + S'(\theta^{(0)})(\theta - \theta^{(0)}) \\ &= S(\theta^{(0)}) - I(\theta^{(0)})(\theta - \theta^{(0)}). \end{aligned}$$

- Solving for $g(\theta) = 0$ yields the improved estimate

$$\theta^{(1)} = \theta^{(0)} + I^{-1}(\theta^{(0)})S(\theta^{(0)}).$$

- Iterate until the estimate converges.

Sketch example

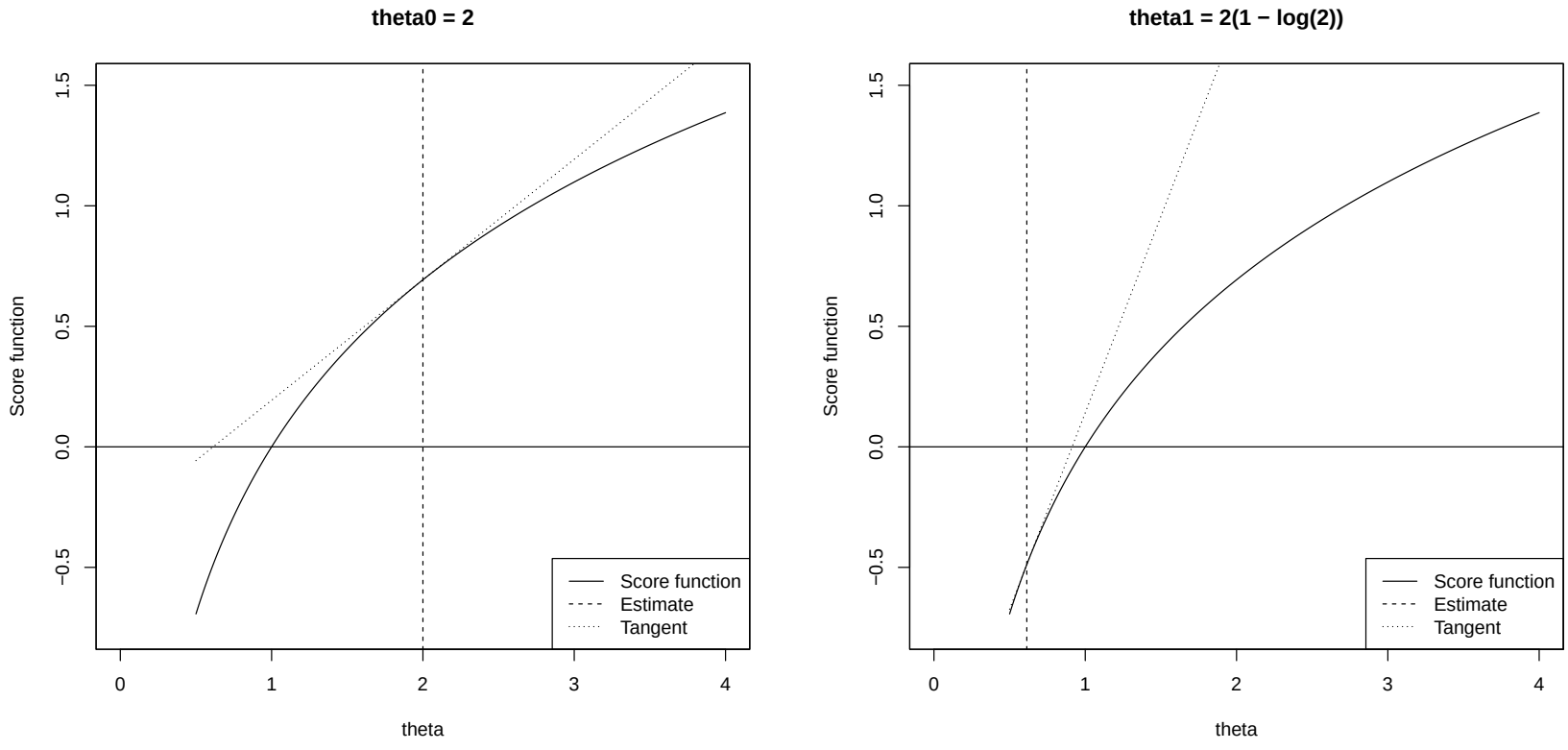


Figure: First two iterations of Newton's method.

Bayes' theorem

Bayes' theorem

- Let E and H be events (*evidence* and *hypothesis*).
- Then

$$\begin{aligned}\mathbb{P}(H|E) &= \frac{\mathbb{P}(E|H)\mathbb{P}(H)}{\mathbb{P}(E)} \\ &= \frac{\mathbb{P}(E|H)\mathbb{P}(H)}{\mathbb{P}(E|H)\mathbb{P}(H) + \mathbb{P}(E|H^c)\mathbb{P}(H^c)}.\end{aligned}$$

- Bayes' theorem generalises to more hypotheses:

$$\mathbb{P}(H_j|E) = \frac{\mathbb{P}(E|H_j)\mathbb{P}(H_j)}{\sum_{k=1}^{n} \mathbb{P}(E|H_k)\mathbb{P}(H_k)}.$$

Example: screening test

- A rare, asymptomatic illness affects 1% of the population.
- A test correctly detects the illness (true positive) with 99% accuracy.
- The same test has a false positive probability of 10%.
- Your test is positive; what is the probability you're ill?
- Let D be the event you're ill and T be the event the test is positive. Then

$$\mathbb{P}(D) = 0.01, \qquad \mathbb{P}(T|D) = 0.99, \qquad \mathbb{P}(T|D^c) = 0.1,$$

$$\begin{aligned}\mathbb{P}(D|T) &= \frac{\mathbb{P}(T|D)\mathbb{P}(D)}{\mathbb{P}(T|D)\mathbb{P}(D) + \mathbb{P}(T|D^c)\mathbb{P}(D^c)} \\ &= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.1 \times 0.99} = \frac{1}{11}.\end{aligned}$$

Bayesian statistics: the prior and posterior

Bayesian statistics

- In frequentist statistics, $\theta = (\theta_1, \ldots, \theta_d)$ was an unknown but fixed quantity.
- The Bayesian paradigm models it as a random variable—model parameters are themselves random.
- The prior distribution with density / mass function $p(\theta)$ is the law of the random variable θ.
- The prior distribution reflects beliefs / knowledge about the parameter before any (new) data is collected.
- Bayesian modelling requires the parameter space Θ, the prior $p(\theta)$, and the observation model for data, $p(\mathbf{x}|\theta)$.
- The observation model is the distribution of observable $\mathbf{X}$ given parameter θ, i.e. it is the likelihood $p(\mathbf{x}|\theta) = L(\theta|\mathbf{x})$.

The posterior distribution

- Once data $\mathbf{x}$ is observed, the prior is updated to the posterior distribution via Bayes' theorem:

$$p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{\int p(\mathbf{x}|\theta')p(\theta')\mathrm{d}\theta'} = \frac{p(\mathbf{x}|\theta)p(\theta)}{Z}.$$

- The right-hand side comprises of the prior $p(\theta)$, the likelihood $p(\mathbf{x}|\theta)$, and a normalising constant which does not depend on θ but ensures $p(\theta|\mathbf{x})$ integrates to one.
- We need conditions on the prior and the likelihood to ensure

$$Z = \int p(\mathbf{x}|\theta')p(\theta')\mathrm{d}\theta' < \infty.$$

Example: Poisson model revisited

- Consider an i.i.d. sample $X_1, \ldots, X_n$ with $X_i \sim \text{Pois}(\lambda)$.
- The likelihood is

$$p(\mathbf{x}|\lambda) = \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}}{\prod_{i=1}^{n} x_i!} \lambda^{\sum_{i=1}^{n} x_i}.$$

- We choose a Gamma distribution as a prior for the mean, i.e. $\lambda \sim G(a, b)$ with density

$$p(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}.$$

Example: Poisson model revisited

- We can compute the posterior as

$$\begin{aligned} p(\lambda|\mathbf{x}) &\propto p(\mathbf{x}|\lambda)p(\lambda) \\ &= \lambda^{\sum_{i=1}^n x_i} e^{-n\lambda} \lambda^{a-1} e^{-b\lambda} \\ &= \lambda^{a-1+\sum_{i=1}^n x_i} e^{-(n+b)\lambda}. \end{aligned}$$

- The right-hand side is the kernel of a Gamma-distribution,

$$\lambda|\mathbf{x} \sim G\left(a + \sum_{i=1}^n x_i, b + n\right).$$

- Because we've identified the kernel, we can read off the normalising constant as well:

$$Z = \frac{(b+n)^{a+\sum_{i=1}^n x_i}}{\Gamma\left(a + \sum_{i=1}^n x_i\right)}.$$

Conjugacy

- A family of priors (e.g. Gaussian, Gamma, Beta, ...) is conjugate to a likelihood $p(\mathbf{x}|\theta)$ if the posterior $p(\theta|\mathbf{x})$ belongs to the same family, with updated parameters.
- In the previous example, the Gamma-prior was conjugate to the Poisson likelihood.
- We say that Gamma-priors and Poisson-likelihoods are a conjugate pair.
- Non-conjugate posteriors are very hard to compute.

Improper and Jeffrey's priors

Example: Gaussian mean

- Consider i.i.d. $X_1, \ldots, X_n$ with $X_i \sim N(\mu, \sigma^2)$ with known σ^2 and unknown μ.
- We set a Gaussian prior: $\mu \sim N(a, b^2)$.
- A completion of the square shows the posterior is Gaussian:

$$p(\mu|\mathbf{x}) \propto \exp\left(\frac{-1}{2b^2}(\mu - a)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

$$\propto \exp\left(\frac{-1}{2\frac{1}{n/\sigma^2+1/b^2}}\left[\mu - \frac{1}{\frac{n}{\sigma^2}+\frac{1}{b^2}}\left(\frac{a}{b^2}+\frac{n\bar{x}}{\sigma^2}\right)\right]^2\right), \text{or}$$

$$\mu|\mathbf{x} \sim N\left(\frac{1}{\frac{n}{\sigma^2}+\frac{1}{b^2}}\left(\frac{a}{b^2}+\frac{n\bar{x}}{\sigma^2}\right), \frac{1}{\frac{n}{\sigma^2}+\frac{1}{b^2}}\right).$$

Improper priors

$$\mu|\mathbf{x} \sim N\left(\frac{1}{\frac{n}{\sigma^2}+\frac{1}{b^2}}\left(\frac{a}{b^2}+\frac{n\bar{x}}{\sigma^2}\right), \frac{1}{\frac{n}{\sigma^2}+\frac{1}{b^2}}\right).$$

- The prior parameters a, b could be specified based on previous studies.
- If little information is available, it is natural to take $b \to \infty$.
- Then the prior is $p(\mu) \propto 1$ which isn't a valid probability density because it has infinite integral. We call the prior improper.
- An improper prior can lead to a proper posterior, in which case Bayesian inference remains valid. Here

$$\mu|\mathbf{x} \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

Example: logit model

- There are many ways to end up with improper priors, e.g. choosing $p(\theta) \propto 1/\theta$.
- For a more involved example, consider the logit model

$$Y_i \sim \text{Bin}(1, \pi_i), \quad \pi_i = \frac{e^{\eta_i}}{e^{\eta_i} + 1}, \quad i \in \{1, \ldots, n\},$$

  where

$$\eta_i = \beta_0 + \beta_1 x_i$$

  is the linear predictor for covariate x_i.
- The likelihood for $\beta = (\beta_0, \beta_1)$ is

$$p(\mathbf{y}|\beta) = \prod_{i=1}^{n} \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

Improper posteriors

- We choose a Gaussian prior for β,

$$\beta \sim N(\mathbf{m}, \mathbf{M})$$

  where $\mathbf{m}$ is the prior mean vector and $\mathbf{M}$ the prior covariance.
- Sending the diagonal elements of $\mathbf{M}$ to infinity, we obtain

$$p(\beta|\mathbf{y}) \propto p(\beta)p(\mathbf{y}|\beta) = 1 \times \prod_{i=1}^{n} \left(\frac{e^{\eta_i}}{e^{\eta_i}+1}\right)^{y_i} \left(\frac{1}{e^{\eta_i}+1}\right)^{1-y_i}$$

$$= \frac{e^{\sum_{i=1}^{n}(\beta_0+\beta_1 x_i)y_i}}{\prod_{i=1}^{n}(e^{\beta_0+\beta_1 x_i}+1)}.$$

- The integral of the posterior over (β_0, β_1) can be infinite (consider e.g. $n = 1$, $y_1 = 0$ for the simplest case) whereupon Bayesian inference is invalid.

Example: binomial model

- Let $X \sim \text{Bin}(n, \pi)$, so that

$$p(x|\pi) \propto \pi^x (1-\pi)^{n-x}.$$

- A non-informative prior is $\pi \sim U(0, 1)$.
- If instead we parametrise by odds, $X \sim \text{Bin}(n, o)$, we get

$$p(x|o) \propto \left(\frac{o}{1+o}\right)^x \left(\frac{1}{1+o}\right)^{n-x}.$$

- Now a non-informative prior is improper: $o \sim U(0, \infty)$.
- *Specification of the prior depends on the parametrisation of the model.*

Jeffrey's prior

- Let $J(\theta) = \mathbb{E}_\theta[I(\theta)] = -\mathbb{E}_\theta[\ell''(\theta|\mathbf{X})]$.
- The Jeffrey's prior is defined as

$$p(\theta) \propto \sqrt{J(\theta)},$$

  which may be proper or improper.
- It is invariant to reparametrisation: if $\phi = h(\theta)$ is a one-to-one map, then

$$p(\phi) \propto \sqrt{J(\phi)}.$$

- For parameter vectors $J(\theta)$ is a matrix and the Jeffrey's prior is

$$p(\theta) \propto \sqrt{|J(\theta)|}.$$

Proof of invariance

- Since h is one-to-one, we can write the likelihood of $\theta = h^{-1}(\phi)$ as
$$\ell(\theta|\mathbf{x}) = \ell(h^{-1}(\phi)|\mathbf{x}).$$
- By the chain rule,
$$\frac{\partial}{\partial\theta}\ell(h^{-1}(\phi)|\mathbf{x}) = \ell'(h^{-1}(\phi)|\mathbf{x})\frac{\mathrm{d}\theta}{\mathrm{d}\phi}.$$
- By the product rule,
$$\frac{\partial^2}{\partial\theta^2}\ell(h^{-1}(\phi)|\mathbf{x}) = \ell''(h^{-1}(\phi)|\mathbf{x})\left(\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right)^2 + \ell'(h^{-1}(\phi)|\mathbf{x})\frac{\mathrm{d}^2\theta}{\mathrm{d}\phi^2}.$$

Proof of invariance

$$\frac{\partial^2}{\partial\theta^2}\ell(h^{-1}(\phi)|\mathbf{x}) = \ell''(h^{-1}(\phi)|\mathbf{x})\left(\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right)^2 + \ell'(h^{-1}(\phi)|\mathbf{x})\frac{\mathrm{d}^2\theta}{\mathrm{d}\phi^2}$$

- Exchanging derivative and integral shows that the second term has expectation zero:

$$\mathbb{E}_\theta\left[\frac{\mathrm{d}^2\theta}{\mathrm{d}\phi^2}\ell'(\theta|\mathbf{X})\right] = \left(\frac{\mathrm{d}^2\theta}{\mathrm{d}\phi^2}\right)\int\frac{L'(\theta|\mathbf{x})}{L(\theta|\mathbf{x})}L(\theta|\mathbf{x})\mathrm{d}\mathbf{x}$$
$$= \left(\frac{\mathrm{d}^2\theta}{\mathrm{d}\phi^2}\right)\frac{\partial}{\partial\theta}\int L(\theta|\mathbf{x})\mathrm{d}\mathbf{x} \propto \frac{\partial}{\partial\theta}1 = 0.$$

- Hence $J(\phi) = \left(\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right)^2 J(\theta)$ and

$$p(\phi) = p(\theta)\left|\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right| \propto \sqrt{J(\theta)}\left|\frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right| = \sqrt{J(\phi)}.$$

Example Jeffrey's priors

- We saw that for an i.i.d. Poisson model, $\ell''(\lambda|\mathbf{x}) = -n\bar{x}/\lambda^2$.
- To derive the Jeffrey's prior, we compute

$$p(\lambda) \propto \sqrt{\mathbb{E}_\lambda[-\ell''(\lambda|\mathbf{X})]} = \sqrt{\frac{n}{\lambda^2}\mathbb{E}_\lambda[X]} = \frac{1}{\sqrt{\lambda}}.$$

- Note that this prior is improper.
- If $Y \sim \text{Bin}(n, \pi)$, then the second derivative of the log-likelihood is

$$\ell''(\pi|y) = -\frac{y}{\pi^2} - \frac{n-y}{(1-\pi)^2}.$$

- Using $\mathbb{E}_\pi[Y] = n\pi$, the (proper) Jeffrey's prior is

$$p(\pi) \propto \sqrt{\mathbb{E}_\pi[-\ell(\pi|Y)]} = \sqrt{\frac{n}{\pi} + \frac{n(1-\pi)}{(1-\pi)^2}} = \sqrt{\frac{n}{\pi(1-\pi)}}.$$

Bayesian point and interval estimators

Bayesian point estimators

- The posterior mean given data $\mathbf{x}$ is given by

$$\hat{\theta}_2 = \mathbb{E}[\theta|\mathbf{x}] = \int \theta p(\theta|\mathbf{x}) \mathrm{d}\theta.$$

- The posterior mode is the maximiser of the posterior density:

$$\hat{\theta}_\infty = \mathrm{argmax}_\theta p(\theta|\mathbf{x}) = \mathrm{argmax}_\theta p(\mathbf{x}|\theta)p(\theta).$$

- The posterior median, $\widehat{\theta}_1$, is the median of the posterior distribution.

Loss functions and risks

- A loss function measures the error between an estimator and the truth, e.g.
  - L^2 loss, $L(\theta, \hat{\theta}) := (\hat{\theta} - \theta)^2$,
  - L^1 loss, $L(\theta, \hat{\theta}) := |\hat{\theta} - \theta|$,
  - zero-one loss, $L(\theta, \hat{\theta}) := \mathbb{1}\{\hat{\theta} \neq \theta\}$,
- The risk is the expected value of the loss:

$$R(\hat{\theta}) = \mathbb{E}[L(\theta, \hat{\theta})|\mathbf{x}] = \int L(\theta, \hat{\theta}) p(\theta|\mathbf{x}) \mathrm{d}\theta.$$

- Note that the randomness is in the parameter θ.

Bayes estimators as risk minimisers

- The posterior mean minimises L^2 risk:

$$\frac{\mathrm{d}}{\mathrm{d}\hat{\theta}}\mathbb{E}[(\hat{\theta}-\theta)^2|\mathbf{x}] = 2\mathbb{E}[\hat{\theta}-\theta|\mathbf{x}] = 0$$
$$\Rightarrow \hat{\theta} = \mathbb{E}[\theta|\mathbf{x}].$$

- The posterior mode minimises zero-one risk.
- The posterior median minimises L^1 risk. By Leibniz's rule,

$$\frac{\mathrm{d}}{\mathrm{d}\hat{\theta}}\mathbb{E}[|\hat{\theta}-\theta||\mathbf{x}]$$
$$= \frac{\mathrm{d}}{\mathrm{d}\hat{\theta}}\int_{-\infty}^{\hat{\theta}}(\hat{\theta}-\theta)p(\theta|\mathbf{x})\mathrm{d}\theta + \frac{\mathrm{d}}{\mathrm{d}\hat{\theta}}\int_{\hat{\theta}}^{\infty}(\theta-\hat{\theta})p(\theta|\mathbf{x})\mathrm{d}\theta = 0$$
$$\Rightarrow \int_{-\infty}^{\hat{\theta}} p(\theta|\mathbf{x})\mathrm{d}\theta = \int_{\hat{\theta}}^{\infty} p(\theta|\mathbf{x})\mathrm{d}\theta.$$

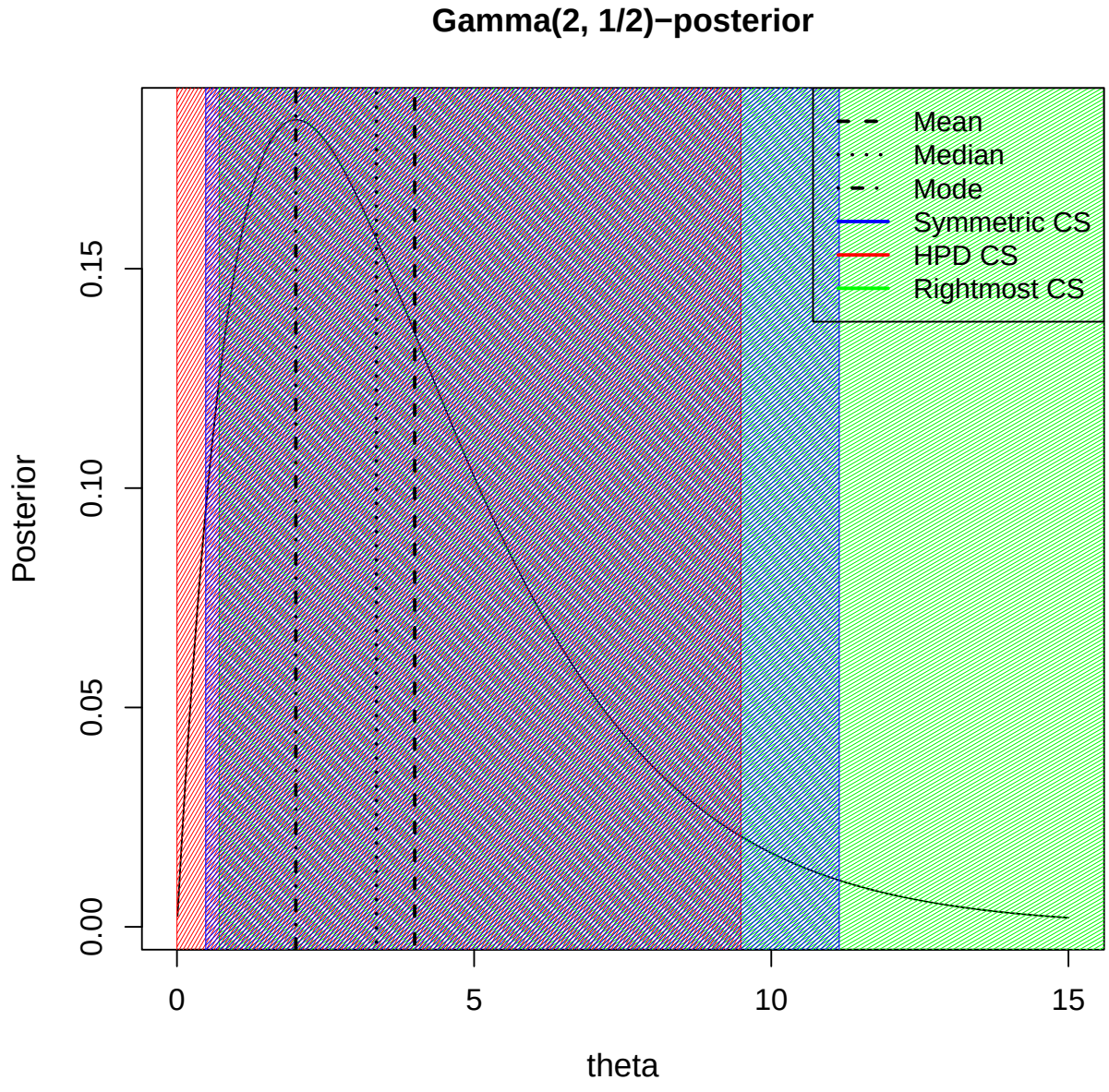
Credible sets

- A set $A \subset \Theta$ is a $1 - \alpha$ credible set if

$$\mathbb{P}(\theta \in A|\mathbf{x}) = p(A|\mathbf{x}) = 1 - \alpha.$$

- This definition is highly non-unique: there are typically uncountably many sets with a given probability mass.
- The highest posterior density (HDP) region is the credible set with given α of least volume.
- Note the difference from confidence intervals. The parameter θ is random, so probabilistic statements about it make sense.

Credible sets



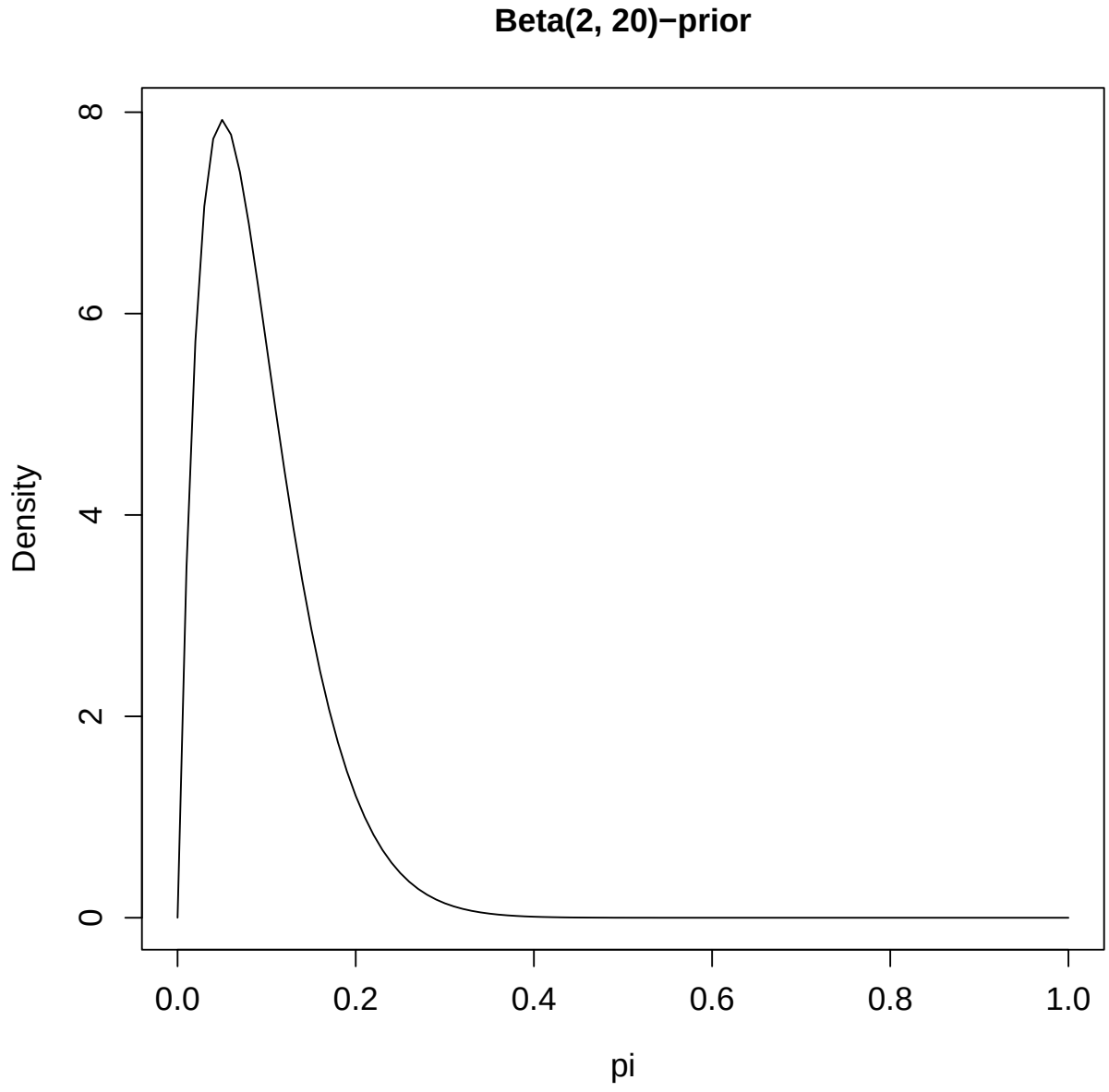
Modelling a rare disease

- Sample size $n = 20$ with $y = 0$ observed cases.
- Observation model: $Y|\pi \sim \text{Bin}(n, \pi)$.
- Interested in $p(\pi|0) = \mathbb{P}(\pi|Y = 0)$.
- Reasonable prior on $[0, 1]$ is $\text{Beta}(a, b)$:

$$p(\pi) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\pi^{a-1}(1-\pi)^{b-1},$$

  with prior mean $\mathbb{E}[\pi] = a/(a+b)$.
- Typical illness rates range from essentially zero to around 0.2, so a reasonable prior is $\pi \sim \text{Beta}(2, 20)$.

Prior density

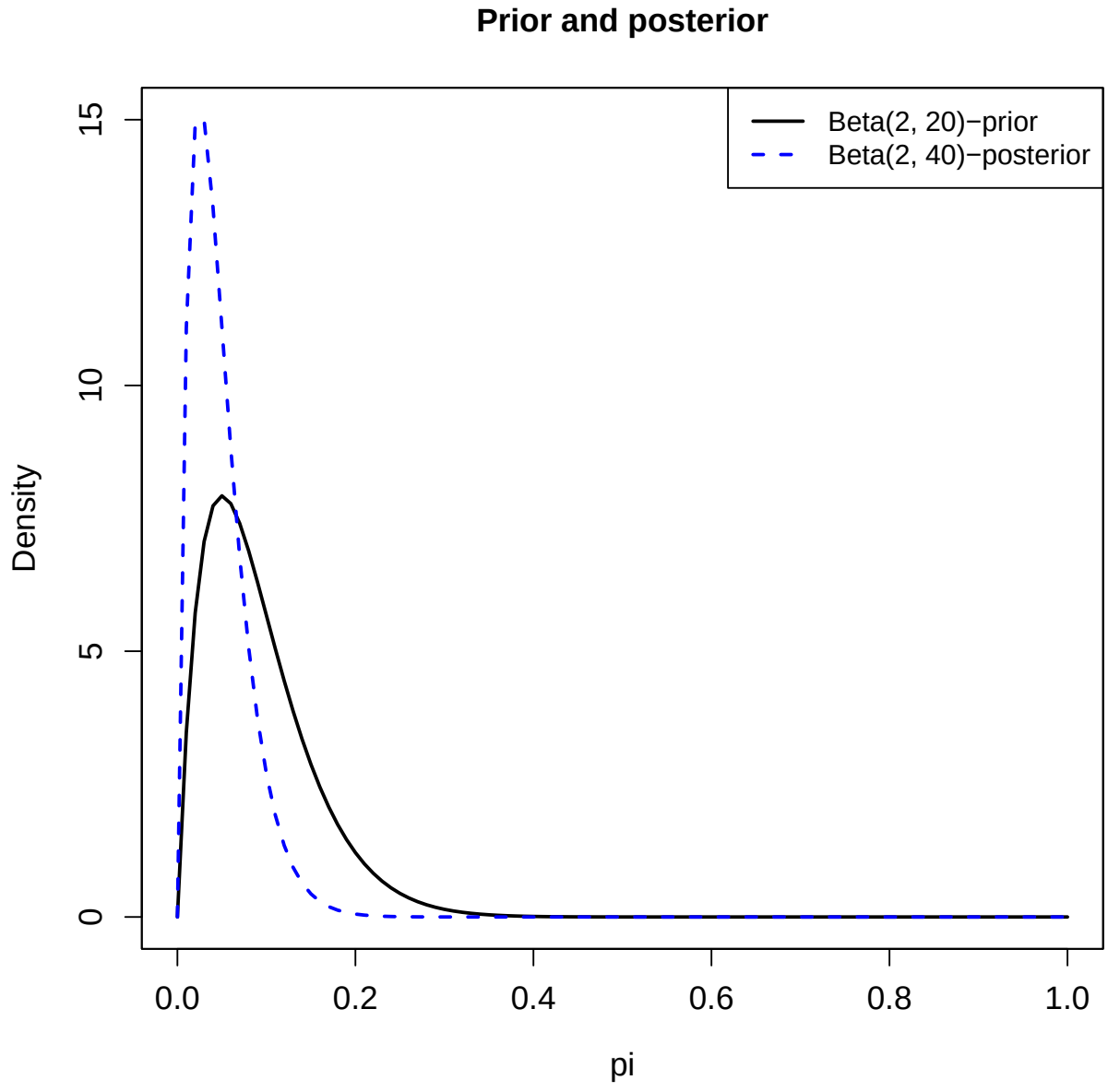


Rare disease posterior

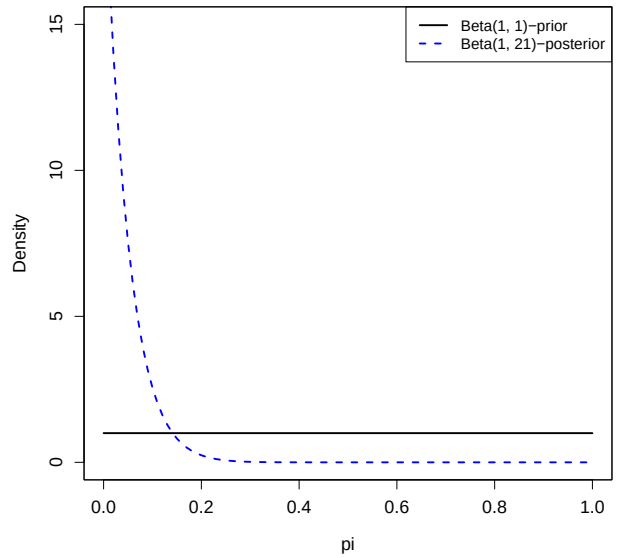
$$p(\pi|y) \propto p(y|\pi)p(\pi) \propto \pi^{y}(1-\pi)^{n-y} \times \pi^{a-1}(1-\pi)^{b-1}$$
$$= \pi^{y+a-1}(1-\pi)^{n+b-y-1}.$$

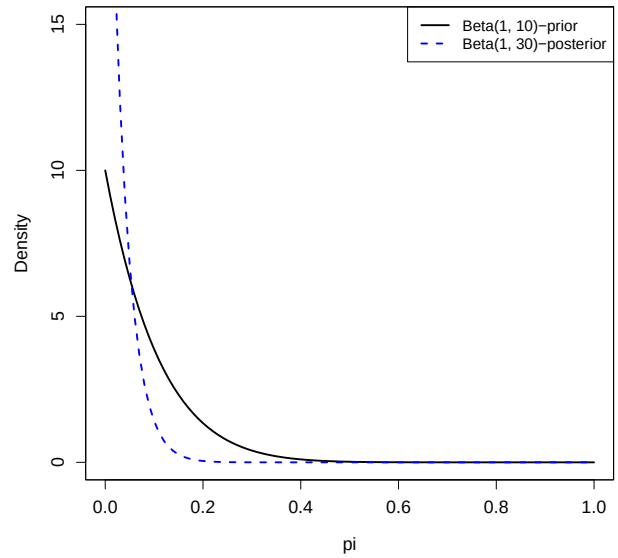
- Posterior $\pi|y \sim \text{Beta}(y+a, n+b-y)$.
- The Beta-prior and binomial likelihood are conjugate.
- The Beta(a, b) prior amounts to adding $a-1$ successes and $b-1$ failures from $a+b-2$ pseudo-observations.

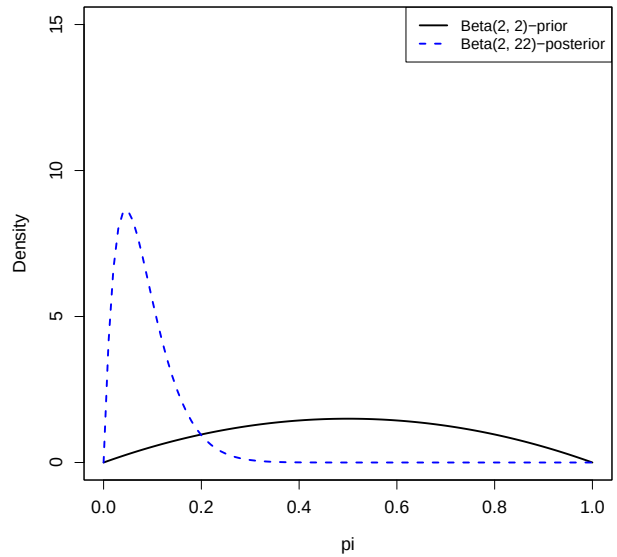
Prior and posterior densities

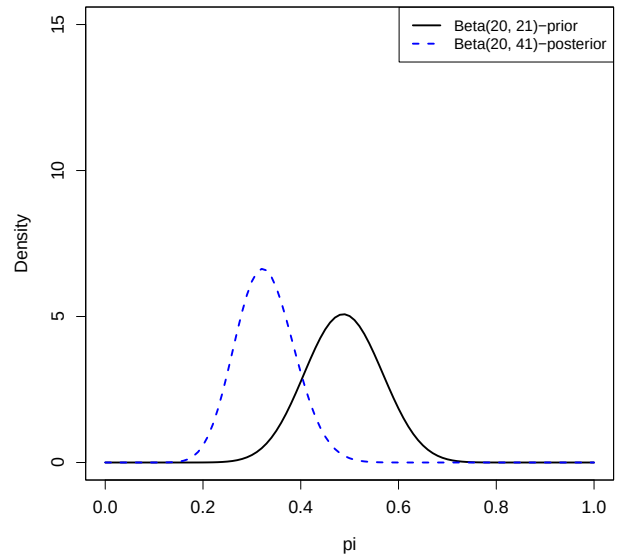


Sensitivity analysis









Point estimates

- Posterior mean

$$\begin{aligned}\mathbb{E}[\pi|y] &= \frac{a+y}{a+b+n} = \frac{a}{a+b+n} + \frac{y}{a+b+n} \\ &= \frac{a+b}{a+b+n}\frac{a}{a+b} + \frac{n}{a+b+n}\frac{y}{n} \\ &= \frac{a+b}{a+b+n}\mathbb{E}[\pi] + \frac{n}{a+b+n}\frac{y}{n}.\end{aligned}$$

- $\mathbb{E}[\pi|y] \sim y/n$ as $n \to \infty$.
- $\mathbb{E}[\pi|y] \sim \mathbb{E}[\pi]$ when $n = 0$.
- Similar for posterior mode:

$$\begin{aligned}\frac{a+y-1}{a+b+n-2} &= \frac{a-1}{a+b+n-2} + \frac{y}{a+b+n-2} \\ &= \frac{a+b-2}{a+b+n-2}\frac{a-1}{a+b-2} + \frac{n}{a+b+n-2}\frac{y}{n}.\end{aligned}$$

The posterior predictive distribution

The posterior predictive distributon

- The prior predictive distribution is

$$\mathbb{P}(X \in A) = \int_{\Theta} p(A|\theta)p(\theta)\mathrm{d}\theta.$$

- The posterior predictive distribution is

$$\mathbb{P}(X \in A|\mathbf{x}) = \int_{\Theta} p(A|\theta)p(\theta|\mathbf{x})\mathrm{d}\theta.$$

- A $1-\alpha$ prediction set is any set $A \subset \Theta$ such that

$$\mathbb{P}(X \in A|\mathbf{x}) = 1-\alpha.$$

Rare disease example continued

- Let Y' denote the disease status of an untested person.
- The posterior predictive distribution of $Y'|y$ is

$$\begin{aligned}\mathbb{P}(Y'=1|y) &= \int_0^1 \mathbb{P}_\pi(Y'=1)p(\pi|y)\mathrm{d}\pi \\ &= \int_0^1 \pi p(\pi|y)\mathrm{d}\pi = \mathbb{E}[\pi|y] \\ &= \frac{a+y}{a+b+n}.\end{aligned}$$

A worked example

Birdwatching

- An MSc Environmental Systems Science student wants to know the relative frequencies of different bird species in the Graz area.
- They have collected pictures from wildlife cameras over the last year, and recorded the species of each observed bird.
- The data are arranged into a vector $\mathbf{n} = (n_1, \ldots n_N)$, whose length is roughly the number of bird species around Graz (in fact, $N \approx 150$)[1]. Entry n_i is the number of times species i was sighted in the cameras.
- We model the underlying frequencies of species as an unknown probability mass function $\mathbf{x} = (x_1, \ldots x_N)$, where $x_i \geq 0$ and $x_1 + \ldots + x_N = 1$.

[1] `https://avibase.bsc-eoc.org/checklist.jsp?region=ATstgz`

The model and likelihood

- We treat the number of birds observed as fixed and denote it by $m = n_1 + \ldots + n_N$. In a more sophisticated analysis, it should also be a variable which depends on the total number of birds in the area.
- We also treat each observed bird as a uniform draw from the underlying population frequencies $\mathbf{x}$.
- The likelihood and log-likelihood of the unknown population frequencies $\mathbf{x}$ given observed counts $\mathbf{n}$ is

$$L(\mathbf{n}|\mathbf{x}) = \binom{m}{n_1, \ldots, n_N} \prod_{i=1}^{N} x_i^{n_i} \propto \left(1 - \sum_{i=1}^{N-1} x_i\right)^{n_N} \prod_{i=1}^{N-1} x_i^{n_i},$$

$$\ell(\mathbf{n}|\mathbf{x}) = C + \sum_{i=1}^{N-1} n_i \log(x_i) + n_N \log\left(1 - \sum_{i=1}^{N-1} x_i\right).$$

Frequentist analysis

- First-order conditions are

$$\frac{\partial}{\partial x_i}\ell(\mathbf{n}|\mathbf{x}) = \frac{n_i}{x_i} - \frac{n_N}{1-\sum_{j=1}^{N-1} x_j} = \frac{n_i}{x_i} - \frac{n_N}{x_N} = 0,$$

  which are solved by the MLEs $\hat{x}_i = n_i/m$.

- Second-order derivatives are

$$\frac{\partial^2}{\partial x_i^2}\ell(\mathbf{n}|\mathbf{x}) = \frac{-n_i}{x_i^2} - \frac{n_N}{(1-\sum_{j=1}^{N} x_j)^2} = \frac{-n_i}{x_i^2} - \frac{n_N}{x_N^2},$$

$$\frac{\partial^2}{\partial x_i \partial x_j}\ell(\mathbf{n}|\mathbf{x}) = \frac{-n_N}{(1-\sum_{j=1}^{N} x_j)^2} = \frac{-n_N}{x_N^2}.$$

Fisher information and asymptotics

- Substituting the MLEs into the second derivatives yields the observed Fisher information:

$$I(\mathbf{x})_{ii} = \frac{m^2}{n_i} + \frac{m^2}{n_N},$$
$$I(\mathbf{x})_{ij} = \frac{m^2}{n_N}.$$

- Its inverse can be computed via the Sherman–Morrison formula, though you would not be expected to know that without looking it up:

$$I^{-1}(\mathbf{x})_{ii} = \frac{n_i(m - n_i)}{m^3},$$
$$I^{-1}(\mathbf{x})_{ij} = \frac{-n_i n_j}{m^3}.$$

- Hence, under regularity conditions (e.g. that no $n_i = 0$),

$$\hat{\mathbf{x}} \sim N(\mathbf{x}, I^{-1}(\hat{\mathbf{x}})),$$

which can be used to approximate confidence ellipsoids etc.

Frequentist R implementation

```
library(expm)
library(ellipse)

ntrials = 100
x = c(0.4, 0.25, 0.35)
n = rmultinom(1, ntrials, x)

mle = n / ntrials
ofi = ntrials^2 * (matrix(rep(1 / n[3], 4), nrow
    = 2) + diag(1/n[1:2]))
ofi_inv = solve(ofi)

ell_coords = ellipse(ofi_inv, center = mle,
   level = 0.95)
plot(ell_coords, type = "l")
points(mle[1], mle[2])
```

Bayesian model

- Recall the likelihood

$$L(\mathbf{n}|\mathbf{x}) = \binom{m}{n_1, \ldots, n_N} \prod_{i=1}^{N} x_i^{n_i} \propto \left(1 - \sum_{i=1}^{N-1} x_i\right)^{n_N} \prod_{i=1}^{N-1} x_i^{n_i}.$$

- It is conjugate to the Dirichlet$(a_1, \ldots, a_N)$ prior,

$$p(\mathbf{x}) \propto \left(1 - \sum_{i=1}^{N-1} x_i\right)^{a_N - 1} \prod_{i=1}^{N-1} x_i^{a_i - 1},$$

  where the $a_i > 0$ variables are hyperparameters.
- The Dirichlet distribution is the multivariate analogue of the Beta-distribution.

Bayesian posterior

- The posterior is

$$p(\mathbf{x}|\mathbf{n}) \propto \left(1 - \sum_{i=1}^{N-1} x_i\right)^{n_N + a_N - 1} \prod_{i=1}^{N-1} x_i^{n_i + a_i - 1}, \text{ or}$$

$$\mathbf{x}|\mathbf{n} \sim \text{Dirichlet}(a_1 + n_1, \ldots, a_N + n_N).$$

- We can interpret a_i as a number of "prior observations" of species i.
- There is no problem with $n_i = 0$ provided $a_i > 0$.
- We can read off posterior point estimates, e.g.

$$\text{the mean: } \hat{x}_i^{(2)} = \frac{a_i + n_i}{a_1 + \ldots + a_N + n_1 + \ldots + n_N},$$

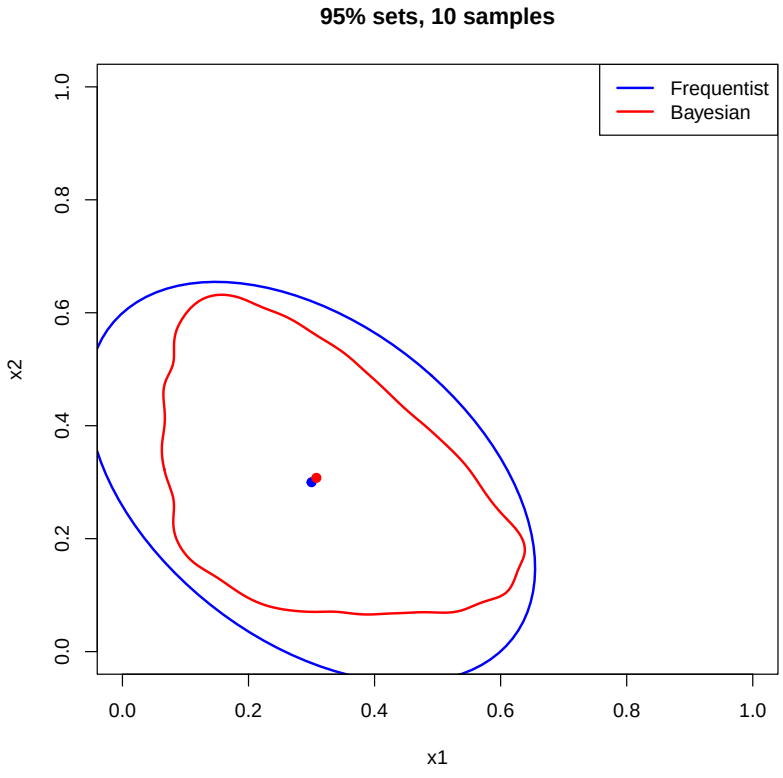
$$\text{the mode: } \hat{x}_i^{(\infty)} = \frac{a_i + n_i - 1}{a_1 + \ldots + a_N + n_1 + \ldots + n_N - N}.$$

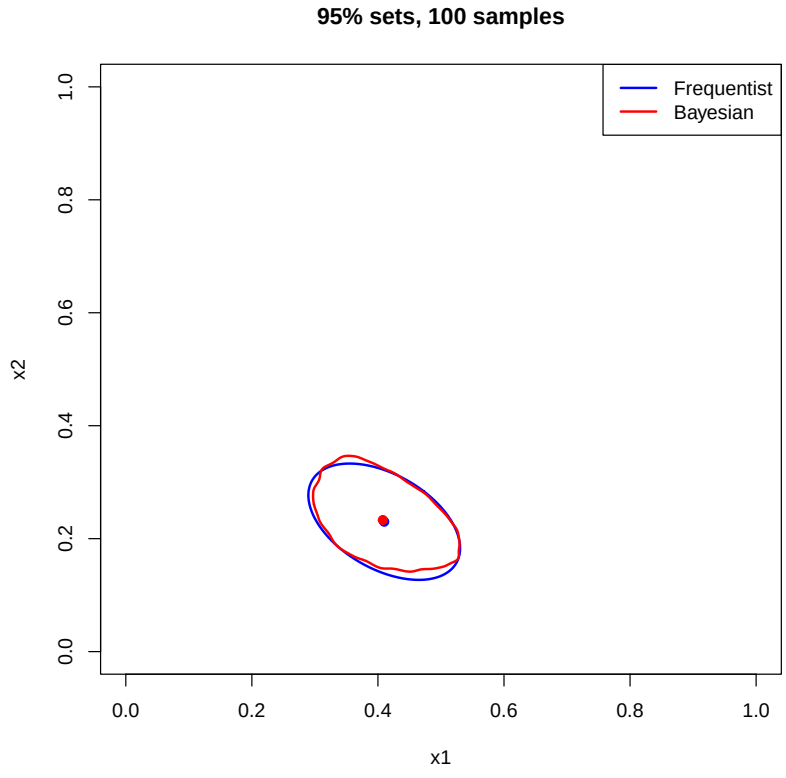
- $a_1 = \ldots = a_N = 1$ is a uniform prior, for which the posterior mode equals the MLE.

Bayesian R implementation

```
library(ks)
ntrials = 100
x = c(0.4, 0.25, 0.35)
n = rmultinom(1, ntrials, x)
a_prior = c(1, 1, 1)
a_post = a_prior + n
npost = 10000
samples = matrix(rep(NA, 3 * npost), nrow=npost)
for (i in 1:npost) {
  samples[i,] = rgamma(3, a_post)
  samples[i,] = samples[i,] / sum(samples[i,])
}
post = kde(samples[,1:2])
cont_95 = contourLines(post$eval.points[[1]],
   post$eval.points[[2]], post$estimate, levels=
   post$cont["5%"])
points(a_post[1] / sum(a_post), a_post[2] / sum(
   a_post))
lines(cont_95[[1]]$x, cont_95[[1]]$y)
```

Frequentist and Bayesian output





Bayesian hypothesis testing

Bayesian hypothesis testing

- Consider two hypotheses, $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_1$.
- Let $p(H_i)$ the the prior probability of hypothesis $i \in \{0, 1\}$.
- The posterior probability of H_i is

$$p(H_i|\mathbf{x}) = \frac{p(\mathbf{x}|H_i)p(H_i)}{p(\mathbf{x})}.$$

- Simple interpretation as conditional probability of a random hypothesis.
- Compare to frequentist case where the p-value is the probability of a similarly extreme result from a replicated experiment under H_0.

Bayes factors

- The posterior odds, or the Bayes factor, are the ratio of posterior probabilities:

$$\mathrm{BF}_{1,0} = \frac{p(H_1|\mathbf{x})}{p(H_0|\mathbf{x})} = \frac{p(\mathbf{x}|H_1)p(H_1)}{p(\mathbf{x}|H_0)p(H_0)}.$$

- The intractable normalising constant $p(\mathbf{x})$ from Bayes theorem cancels. But

$$p(H_i|\mathbf{x}) \propto p(\mathbf{x}|H_i)p(H_1) = \int_{\Theta_i} p(\mathbf{x}|\theta, H_i)p(\theta, H_i)\mathrm{d}\theta$$

  is the normalising constant under H_i:

$$p(\theta|\mathbf{x}, H_i) = \frac{p(\mathbf{x}|\theta, H_i)p(\theta|H_i)}{p(\mathbf{x}|H_i)}.$$

- Hence $p(H_i|\mathbf{x})$ is also intractable in general.

Interpreting Bayes factors

- $p(H_1|\mathbf{x}) = p(H_0|\mathbf{x}) \Rightarrow$ both hypotheses equally plausible.
- Bayes factors are typically reported in applications.
- There are well-established benchmarks for interpreting Bayes factors[2]:

$BF_{1,0}$	Evidence against H_0
1–3	Not worth more than a bare mention
3–20	Positive
20–150	Strong
$>$150	Very strong

- For simple hypotheses, i.e. $\Theta_i = \{\theta_i\}$, the Bayes factor equals the frequentist likelihood ratio.

[2] R. E. Kass & A. E. Raftery. Bayes factors. *J. Am. Stat. Assoc.* 90(430):773–795, 1995.

Example: avalanches in the Alps

- We have a time series of numbers of daily avalanches in the Alps: $\mathbf{x} = (x_1, \ldots, x_n)$.
- Authorities want to know whether the data is overdispersed, i.e. whether the variance exceeds the mean, which is important for e.g. quantifying likelihood of extreme events.
- H_0: no overdispersion, $X_i \sim \text{Pois}(\lambda)$.
- H_1: overdispersion, $X_i \sim \text{NegBin}(r, \pi)$.
- We choose conjugate priors: $\lambda \sim \text{Gamma}(a_0, b_0)$ for H_0 and $\pi \sim \text{Beta}(a_1, b_1)$ for H_1. We assume the r-parameter under H_1 is known from e.g. earlier work or expert consensus.
- We consider both hypotheses equally likely *a priori*.

Avalanche Bayes factor

- Posterior under $H_0 : \lambda|\mathbf{x} \sim \text{Gamma}(a_0 + n\bar{x}, b_0 + n)$.
- Posterior under $H_1 : \pi|\mathbf{x} \sim \text{Beta}(a_1 + rn, b_1 + n\bar{x})$.
- The Bayes factor is the ratio of normalising constants:

$$\text{BF}_{1,0} = \frac{\Gamma(a_1 + rn)\Gamma(b_1 + n\bar{x})}{\Gamma(a_1 + rn + b_1 + n\bar{x})} \cdot \frac{\Gamma(a_0 + n\bar{x})}{(b_0 + n)^{a_0 + n\bar{x}}}.$$

Asymptotic performance guarantees

Frequentist guarantees for Bayesian inference

- Recall that the frequentist MLE is asymptotically Gaussian under standard assumptions:

$$\widehat{\theta}_{\text{MLE}} \sim N(\theta, I^{-1}(\widehat{\theta}_{\text{MLE}}))$$

  as $n \to \infty$, where $I(\widehat{\theta}_{\text{MLE}})$ is the observed Fisher information.
- Consider Bayesian model $(p(\cdot|\theta), \theta \in \Theta)$ with prior $p(\theta)$ on Θ.
- Suppose $X_1, \ldots, X_n \overset{i.i.d.}{\sim} p(\cdot|\theta_0)$ with data-generating parameter $\theta_0 \in \Theta$.
- When does the posterior $p(\cdot|\mathbf{x})$ place more and more mass at (or near) θ_0 as $n \to \infty$?
- Write $p_\theta = p(\cdot|\theta)$ and $p_0 = p(\cdot|\theta_0)$ for brevity.

[2] A. W. van der Vaart. Asymptotic Statistics, Chapter 10. Cambridge University Press, 2012.

Bernstein–von Mises theorem: assumptions

- Suppose the model admits densities $(p_\theta : \theta \in \Theta)$, that the Fisher information matrix $I(\theta_0)$ is nonsingular, and that the prior $p(\theta)$ has a continuous, strictly positive density.
- Suppose the model is differentiable in quadratic mean, i.e. there exists a function f such that, as $\theta \to \theta_0$,

$$\int \frac{(\sqrt{p_\theta(x)} - \sqrt{p_{\theta_0}(x)} - \frac{1}{2}(\theta - \theta_0)^T f(x)\sqrt{p_{\theta_0}(x)})^2}{\|\theta - \theta_0\|^2} \mathrm{d}x \to 0.$$

- Suppose that for every $\varepsilon > 0$ there exists a sequence of hypothesis tests $\phi_n(\mathbf{x}_{1:n}) \in [0, 1]$ of $H_0 : \theta = \theta_0$ vs. $H_1 : \|\theta - \theta_0\| > \varepsilon$, i.e.

$$\lim_{n\to\infty} \mathbb{E}_{\theta_0}[\phi_n(\mathbf{X}_{1:n})] = 0,$$

$$\lim_{n\to\infty} \sup_{\theta:\|\theta-\theta_0\|>\varepsilon} \{\mathbb{E}_\theta[\phi_n(\mathbf{X}_{1:n})]\} = 1.$$

Bernstein–von Mises theorem: conclusion

- Then, for any estimator $\widehat{\theta}_n$ satisfying

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \to N(\theta_0, I^{-1}(\theta_0)),$$

in distribution as $n \to \infty$, the posterior $p(\cdot|\mathbf{x}_{1:n})$ satisfies

$$\|p(\cdot|\mathbf{x}_{1:n}) - N(\widehat{\theta}_n, I(\theta_0)^{-1}/n)\|_{\mathrm{TV}} \to 0$$

in $p(\cdot|\theta_0)$-probability, as $n \to 0$.

- The total variation distance $\|\cdot\|_{\mathrm{TV}}$ is

$$\|p - q\|_{\mathrm{TV}} := \frac{1}{2}\int |p(x) - q(x)\mathrm{d}x = \sup_A |p(A) - q(A)|.$$

Bernstein–von Mises theorem: comments

- The criterion for $\widehat{\theta}_n$ is exactly the asymptotic normality of the MLE, i.e. under standard conditions we can take $\widehat{\theta}_n = \widehat{\theta}_{\mathrm{MLE}}$.
- We can similarly use standard hypothesis tests as candidates for ϕ_n.
- Nonsingularity of $I(\theta_0)$ rules out $\theta_0 \in \partial\Theta$ and requires the model to be identifiable: $\theta \neq \theta' \Rightarrow p_\theta \neq p_{\theta'}$.
- Under the theorem's assumptions, Bayesian credible sets are (asymptotically) valid frequentist confidence sets.
- But the assumptions are very often violated in practice. Most immediately, models are almost never well-posed, i.e. a true data-generating mechanism $\theta_0 \in \Theta$ does not exist.

Bernstein–von Mises example

- Recall that the Jeffrey's prior for $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$ is $p(\lambda) \propto 1/\sqrt{\lambda}$, which is continuous and strictly positive.
- Recall the Fisher information for the Poisson likelihood, $I(\lambda) = n\bar{\mathbf{x}}/\lambda^2$; nonsingular as long as $n\bar{\mathbf{x}} > 0$ and $\lambda > 0$.
- Let $g_\lambda(x) = \sqrt{p_\lambda(x)}$. Then

$$
\begin{aligned}
g'_\lambda(x) &= \frac{1}{2}\frac{p'_\lambda(x)}{\sqrt{p_\lambda(x)}} = \frac{1}{2}\frac{p'_\lambda(x)}{p_\lambda(x)} g_\lambda(x) = \frac{1}{2}\ell'(\lambda|x) g_\lambda(x) \\
&= \frac{x-\lambda}{2\lambda} g_\lambda(x), \\
g''_\lambda(x) &= \frac{g_\lambda(x)}{2}\left(\ell''(\lambda|x) + \frac{1}{2}\ell'(\lambda|x)^2\right) \\
&= \frac{g_\lambda(x)}{4\lambda^2}(x^2 - 2x - 2\lambda x + \lambda^2).
\end{aligned}
$$

Bernstein–von Mises example: Taylor expansion

- By Taylor's theorem with Lagrange-form remainder, there is $\mu \in [\lambda_0, \lambda]$ such that

$$\begin{aligned} g_\lambda(x) - g_{\lambda_0}(x) &= (\lambda - \lambda_0) g'_{\lambda_0}(x) + \frac{(\lambda - \lambda_0)^2}{2} g''_\mu(x) \\ &= \frac{(\lambda - \lambda_0)(x - \lambda_0)}{2\lambda_0} g_{\lambda_0}(x) \\ &\quad + \frac{(\lambda - \lambda_0)^2 (x^2 - 2x - 2\mu x + \mu^2)}{8\mu^2} g_\mu(x). \end{aligned}$$

- Hence choosing $f(x) = \ell'(\lambda|x) = (x - \lambda_0)/\lambda_0$ yields

$$\begin{aligned} &\left(\sqrt{p_\lambda(x)} - \sqrt{p_{\lambda_0}(x)} - \frac{1}{2}(\lambda - \lambda_0) f(x) \sqrt{p_{\lambda_0}(x)} \right)^2 \\ &= \frac{(\lambda - \lambda_0)^4 (x^2 - 2x - 2\mu x + \mu^2)^2}{64\mu^4} p_\mu(x). \end{aligned}$$

Bernstein–von Mises: differentiability in quadratic mean

- It remains to show that, as $\lambda \to \lambda_0$,

$$\sum_{x=0}^{\infty} \frac{(\lambda - \lambda_0)^4 (x^2 - 2x - 2\mu x + \mu^2)^2}{64\mu^4 (\lambda - \lambda_0)^2} p_\mu(x) \mathrm{d}x \to 0$$

- The convergence holds because, for $X \sim \text{Pois}(\mu)$ we have $\mathbb{E}[X^j] = O(\mu^j)$ for $j \in \{1, 2, 3, 4\}$, and $(\lambda - \lambda_0)^2$ in the denominator cancels.
- Hence differentiability in quadratic mean holds with $f(x) = \ell'(\lambda | x)$.

Bernstein–von Mises example: hypothesis tests

- For fixed $\mu_0 > 0$ and $\varepsilon > 0$ define

$$\phi_n(\mathbf{x}) = \mathbb{1}\{|\bar{\mathbf{x}} - \mu_0| > \varepsilon\}.$$

- Recall that $\bar{\mathbf{X}} \to \mu_0$ as $n \to \infty$.
- Then, as $n \to \infty$,

$$\mathbb{E}_{\mu_0}[\phi_n(\mathbf{X})] = \mathbb{P}(|\bar{\mathbf{X}} - \mu_0| > \varepsilon) \to 0.$$

- Likewise, for $|\mu - \mu_0| > \varepsilon$,

$$\mathbb{E}_{\mu}[\phi_n(\mathbf{X})] = \mathbb{P}(|\bar{\mathbf{X}} - \mu_0| > \varepsilon) \to \mathbb{1}\{|\mu - \mu_0| > \varepsilon\} = 1.$$

- Since the above holds for all $|\mu - \mu_0| > \varepsilon$, we also have

$$\sup_{\mu:|\mu-\mu_0|>\varepsilon} \{\mathbb{E}_{\mu}[\phi_n(\mathbf{X})]\} \to 1.$$